Pre-Lab 02:
Operational Amplifier Applications

ECE 1212: Electronic Circuit Design Lab

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Team Roles:

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Circuit Construction	Stefan Zhang
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1) Pre-Lab (2A)

1.1) Part I: Instrumentation Amplifier

1.1.1) Question 1: Circuit Analysis

For ideal op amps, the input currents are zero and the two input terminals of each op amp are at approximately the same voltage.

For the upper input stage,

$$\frac{V_{01} - V_1}{R_2} = \frac{V_1 - V_2}{R_1},$$

so

$$V_{01} = V_1 + \frac{R_2}{R_1}(V_1 - V_2).$$

Similarly, for the lower input stage,

$$V_{02} = V_2 + \frac{R_2}{R_1}(V_2 - V_1).$$

Therefore,

$$V_{02} - V_{01} = \left(1 + \frac{2R_2}{R_1}\right)(V_2 - V_1).$$

For the final difference amplifier as drawn in the handout,

$$V_o = \frac{R_4}{R_3}(V_{02} - V_{01}),$$

and hence

$$V_o = \frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_1}\right)(V_2 - V_1).$$

Sign consistency note: the circuit shown in the handout gives a positive coefficient for $(V_2 - V_1)$, while the handout specifies

$$V_o = -\frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_1}\right)(V_2 - V_1).$$

To obtain the required negative gain, the two signal inputs of the final difference-amplifier stage must be interchanged. The resulting transfer function is

$$V_o = -\frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_1}\right)(V_2 - V_1)$$

so that

$$K = -\frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_1}\right).$$

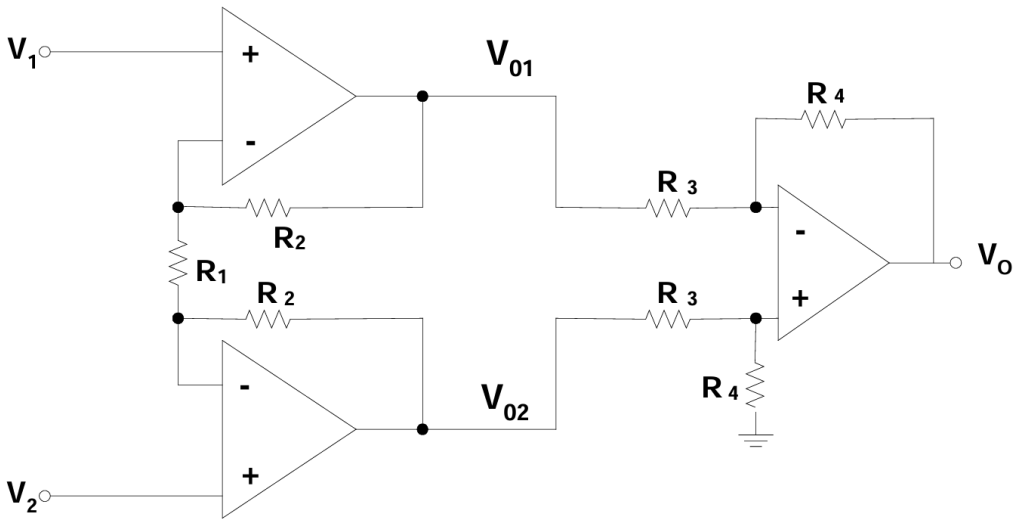


Figure 1: Instrumentation amplifier designed for $K = -3$

1.1.2) Question 2: Design for $K = -3$

Given

$$R_2 = R_3 = R_4 = 1\,\mathrm{k}\Omega,$$

the required gain is

$$-3 = -\frac{R_4}{R_3}\left(1 + \frac{2R_2}{R_1}\right).$$

Since $R_4/R_3 = 1$,

$$3 = 1 + \frac{2(1\,\mathrm{k}\Omega)}{R_1},$$

which gives

$$\boxed{R_1 = 1\,\mathrm{k}\Omega}.$$

Thus, the nominal design is

$$R_1 = R_2 = R_3 = R_4 = 1\,\mathrm{k}\Omega.$$

For $V_1 = 0$ and $V_2 = 1\,\mathrm{V_{pp}}$, the expected output is approximately

$$V_o = 3\,\mathrm{V_{pp}},$$

with an inversion relative to $V_2 - V_1$.

1.2) Part II: Ramp Generation

When the relay contact is open, the circuit operates as an inverting integrator. For an ideal op amp,

$$V_- \approx V_+ = 0.$$

The current through R is

$$i_R = \frac{V_{PS}}{R}.$$

Since the op-amp input current is zero, this current flows through the capacitor:

$$\frac{V_{PS}}{R} = -C\frac{dV_o}{dt}.$$

Therefore,

$$\frac{dV_o}{dt} = -\frac{V_{PS}}{RC}.$$

With the capacitor initially discharged,

$$V_o(0) = 0,$$

and thus

$$V_o(t) = -\frac{V_{PS}}{RC}t.$$

The required output is

$$V_o(0.2\,\mathrm{s}) = -5\,\mathrm{V}.$$

Hence,

$$\frac{V_{PS}}{RC} = \frac{5}{0.2} = 25\,\mathrm{V/s}.$$

Choose

$$\boxed{R = 100\,\mathrm{k\Omega}, \qquad C = 0.10\,\mu\mathrm{F}}.$$

Then

$$RC = (100\,\mathrm{k\Omega})(0.10\,\mu\mathrm{F}) = 0.010\,\mathrm{s},$$

so the required DC source is

$$\boxed{V_{PS} = (25\,\mathrm{V/s})(0.010\,\mathrm{s}) = 0.25\,\mathrm{V}}.$$

The resulting output is

$$V_o(t) = -25t,$$

with V_o in volts and t in seconds. Therefore,

$$V_o(0.2\,\mathrm{s}) = -5\,\mathrm{V}.$$

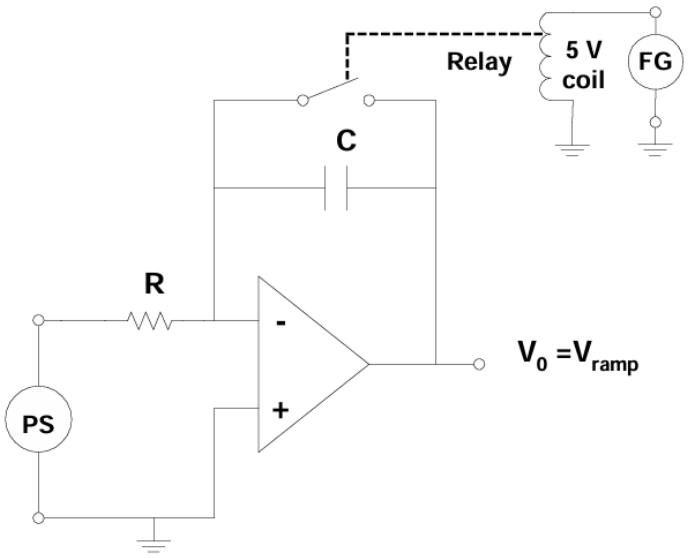


Figure 2: Ramp-generator design using an op-amp integrator and relay

The desired ramp lasts for 200 ms, followed by a 100 ms reset interval. Thus,

$$T = 200\,\text{ms} + 100\,\text{ms} = 300\,\text{ms},$$

and

$$f = \frac{1}{T} \approx 3.33\,\text{Hz}.$$

Using the relay's normally-open (NO) contact across the capacitor:

FG Output	Relay Contact	Circuit State
0 V	Open	Ramp generation
5 V	Closed	Capacitor reset

Therefore, the function generator should be set to a square wave with

$$\boxed{f = 3.33\,\text{Hz}, \qquad V_{\text{high}} = 5\,\text{V}, \qquad V_{\text{low}} = 0, \qquad D = 33.3\%}$$

Equivalently, the voltage setting is approximately $5\,\text{V}_{\text{pp}}$ with a $+2.5\,\text{V}$ DC offset. The relay is energized for 100 ms to reset the capacitor and de-energized for 200 ms to generate the ramp.

If the normally-closed (NC) contact is used instead, the required high-state duty cycle becomes 66.7%.

2) Pre-Lab (2B)

2.1) Part III: Instrumentation Amplifier Simulation

The instrumentation amplifier from Part I will be simulated using three LM741/NS op amps from the NAT_SEMI library. All op amps use $\pm 12\,\mathrm{V}$ power supplies.

The component values are

$$R_1 = R_2 = R_3 = R_4 = 1\,\mathrm{k}\Omega,$$

with the final difference-amplifier inputs connected to obtain

$$V_o = -3(V_2 - V_1).$$

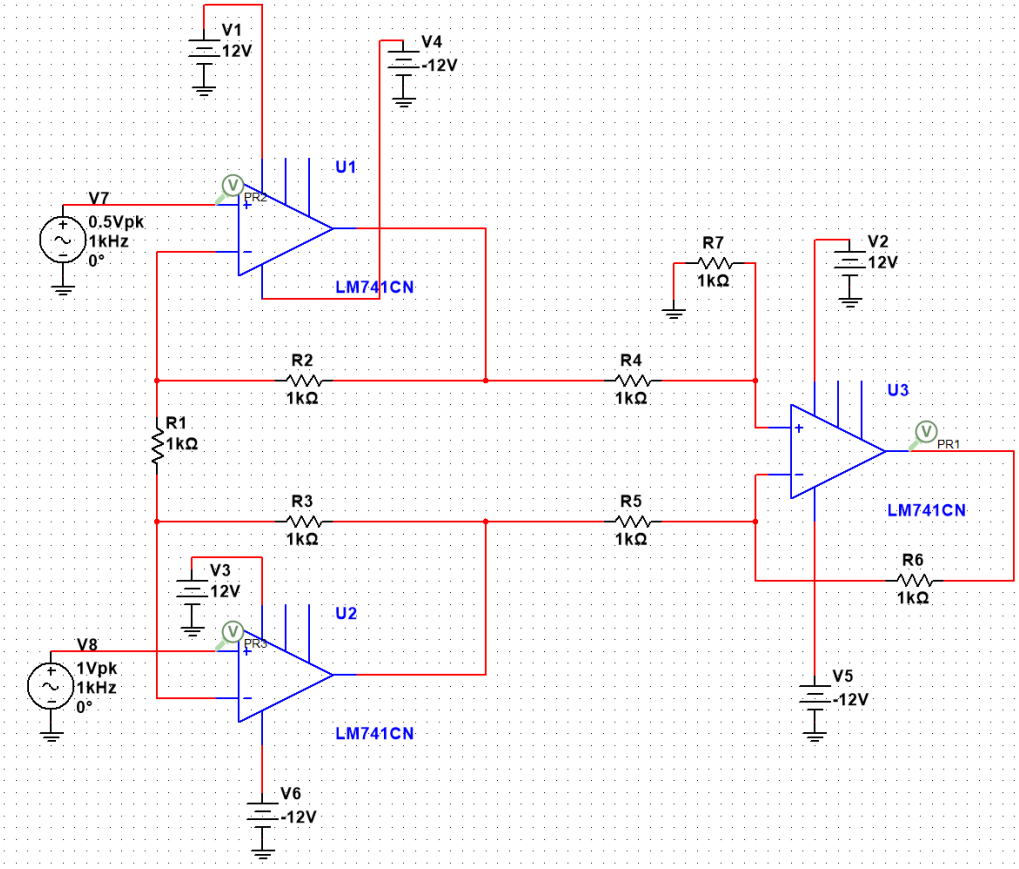


Figure 3: PSpice/Multisim instrumentation-amplifier simulation circuit

For the LM741/NS,

$$\text{Pin } 7 = +12\,\mathrm{V}, \qquad \text{Pin } 4 = -12\,\mathrm{V}.$$

The two simulation cases are:

Table 1: Instrumentation-amplifier simulation inputs and expected outputs

Case	$\mathbf{V_1}$	$\mathbf{V_2}$	**Expected** V_o
Inputs 1	Sine, 1 kHz, 1 V_{pp}	Sine, 1 kHz, 2 V_{pp}	$\approx 3\,V_{pp}$
Inputs 2	Sine, 1 kHz, 1 V_{pp}	Sine, 1 kHz, 1 V_{pp}	$\approx 0\,V$

For Inputs 1, assuming the two sources are synchronized and in phase,

$$V_2 - V_1 = 1\,V_{pp},$$

so

$$V_o = -3(V_2 - V_1)$$

has an expected magnitude of

$$3\,V_{pp}.$$

For Inputs 2, the two synchronized input signals are identical, so

$$V_2 - V_1 = 0$$

and ideally

$$V_o = 0.$$

Small differences from the ideal results may occur because of the non-ideal properties of the LM741 model.